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CIND 119:Introduction to Big Data Analytics Assignment 1

Task 1:

GOAL:

To upload the Heart File onto SAS and print the first 10 observations (rows). To see how many observations and the type of data in each variable proc contents will also be used.

CODE/RESULT:

```
proc import
/* out keyword is used to name a table*/
out=heart
/* Datafile keyword takes the path of the file from the hard disk*/
datafile ='/home/u64430375/Labs CIND119/Assignment 1/heart.csv'
/* dbms= csv replace is telling SAS it is a csv file. */
dbms=csv replace;
/* Getnames=yes will use first line of the csv file as column names*/
getnames=yes;
run;
/* data keyword takes the name of the SAS table imported as auto_csv.
print keyword outputs the contents in Results Viewer */
proc print data=heart(obs=10);
title "Heart Dataset";
run;
/*Show dataset metadata (Total rows and variable types) */
proc contents data=heart;
   title "Dataset Summary: Total Rows and Variables";
run;
```

Heart Dataset

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	63	1	3	145	233	1	0	150	0	2.3	0	0	1	1
2	37	1	2	130	250	0	1	187	0	3.5	0	0	2	1
3	41	0	1	130	204	0	0	172	0	1.4	2	0	2	1
4	56	1	1	120	236	0	1	178	0	0.8	2	0	2	1
5	57	0	0	120	354	0	1	163	1	0.6	2	0	2	1
6	57	1	0	140	192	0	1	148	0	0.4	1	0	1	1
7	56	0	1	140	294	0	0	153	0	1.3	1	0	2	1
8	44	1	1	120	263	0	1	173	0	0	2	0	3	1
9	52	1	2	172	199	1	1	162	0	0.5	2	0	3	1
10	57	1	2	150	168	0	1	174	0	1.6	2	0	2	1

ANALYSIS:

The heart dataset was successfully loaded and there are 303 observations in this dataset. Every column is filled with numeric values.The first 13 columns are attributes

that could potentially determine the 14th column which is labelled target,which indicates if a patient has heart disease. Looking at the first 10 rows, there are a mix of continuous numerical values (like age and chol) and binary categorical values (like sex and exercise induced angina).

Task 2:

GOAL:

To perform a basic data analysis to know the max, min, mean and standard deviation for each variable.

CODE/RESULT:

```
/* Performing basic analysis using proc means*/
proc means data=heart n means std min max;
        title" Summary Statistics for Heart Dataset";
run;
```

Summary Statistics for Heart Dataset

The MEANS Procedure

Variable	N	Mean	Std Dev	Minimum	Maximum
age	303	54.3663366	9.0821010	29.0000000	77.0000000
sex	303	0.6831683	0.4660108	0	1.0000000
cp	303	0.9669967	1.0320525	0	3.0000000
trestbps	303	131.6237624	17.5381428	94.0000000	200.0000000
chol	303	246.2640264	51.8307510	126.0000000	564.0000000
fbs	303	0.1485149	0.3561979	0	1.0000000
restecg	303	0.5280528	0.5258596	0	2.0000000
thalach	303	149.6468647	22.9051611	71.0000000	202.0000000
exang	303	0.3267327	0.4697945	0	1.0000000
oldpeak	303	1.0396040	1.1610750	0	6.2000000
slope	303	1.3993399	0.6162261	0	2.0000000
ca	303	0.7293729	1.0226064	0	4.0000000
thal	303	2.3135314	0.6122765	0	3.0000000
target	303	0.5445545	0.4988348	0	1.0000000

ANALYSIS:

From this table we can see that the participants of the study are between the ages of 29 and 77 with the average participant being 54 years old. The variable sex, has the minimum value is 0 and maximum value is 1 because it is binary. The mean tells us that 68% of patients in the dataset are male.

Task 3:

GOAL:

To standardize the numerical attributes in the dataset.

CODE/RESULT:

```
/* Standardize numerical attributes */
proc stdize data=heart out=heart_std method=std;
   var age trestbps chol thalach oldpeak;
run;

proc print data=heart_std (obs=10);
   title "Standardized Data (First 10 Rows)";
run;
```

Task 3: Standardized Data (First 10 Rows)

Obs	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
1	0.9506240215	1	3	0.7626940758	-0.255910365	1	0	0.0154172814	0	1.0855422911	0	0	1	1
2	-1.912149695	1	2	-0.092584625	0.0720802521	0	1	1.6307737425	0	2.1190672376	0	0	2	1
3	-1.471722969	0	1	-0.092584625	-0.815423771	0	0	0.9758995015	0	0.3103985813	2	0	2	1
4	0.1798772518	1	1	-0.662770426	-0.198029668	0	1	1.2378491979	0	-0.206363892	2	0	2	1
5	0.2899839332	0	0	-0.662770426	2.078611086	0	1	0.5829749569	1	-0.37861805	2	0	2	1
6	0.2899839332	1	0	0.4776011755	-1.046946559	0	1	-0.071899284	0	-0.550872207	1	0	1	1
7	0.1798772518	0	1	0.4776011755	0.9209971433	0	0	0.1463921296	0	0.2242715024	1	0	2	1
8	-1.141402925	1	1	-0.662770426	0.3228966063	0	1	1.0195577842	0	-0.895380523	2	0	3	1
9	-0.260549474	1	2	2.3021957372	-0.911891599	1	1	0.5393166742	0	-0.464745129	2	0	3	1
10	0.2899839332	1	2	1.047786976	-1.509992136	0	1	1.063216067	0	0.482652739	2	0	2	1

ANALYSIS:

The variables age, trestbps, chol, thalach, and oldpeak have been transformed into standardized values (Z-scores). This will help to ensure all variables are on the same scale with a mean of 0 and a standard deviation of 1. Standardization prevents variables with larger numeric ranges like cholesterol from having a greater influence on distance calculations used in k-means clustering.

Task 4: Clustering

GOAL:

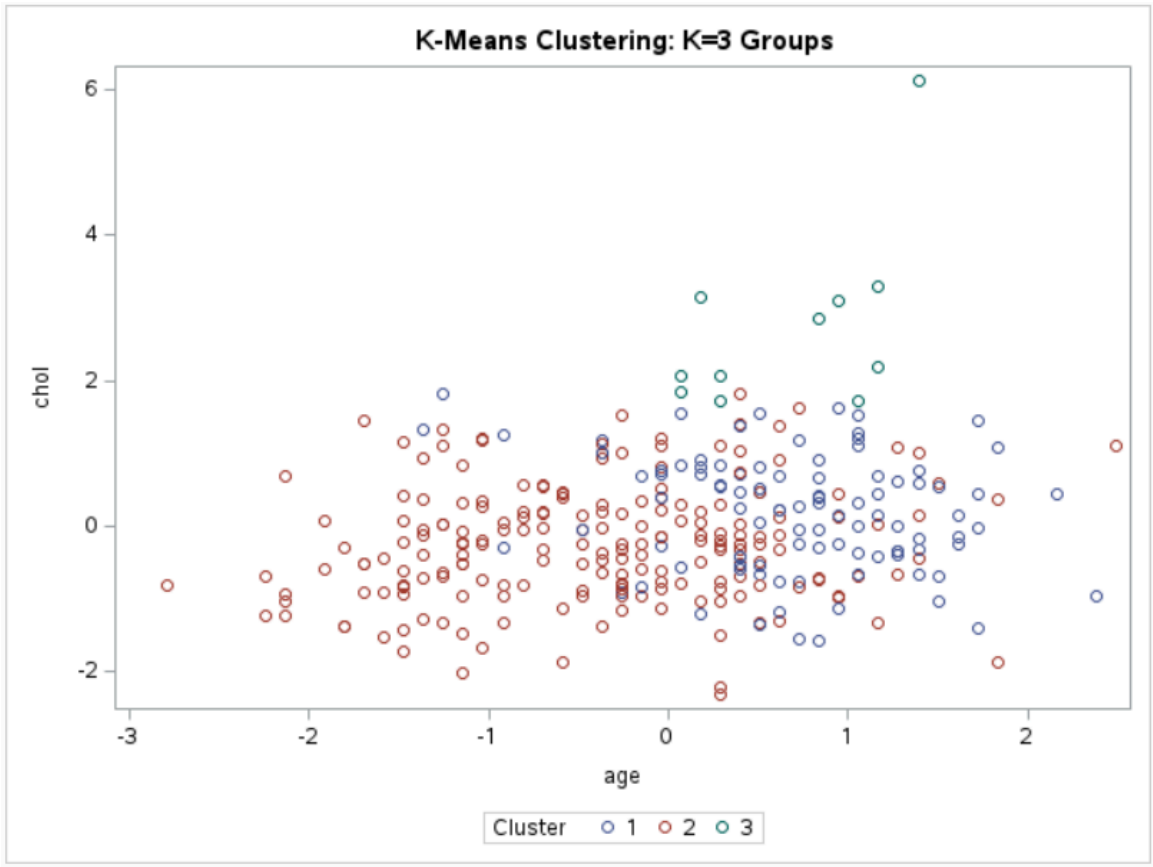
Make clusters with patients using k-means. Compare the original data to the Standardized one and try=2, 3, 4, 5 and evaluate the solutions. Choose the best K value based on the RMS Std.

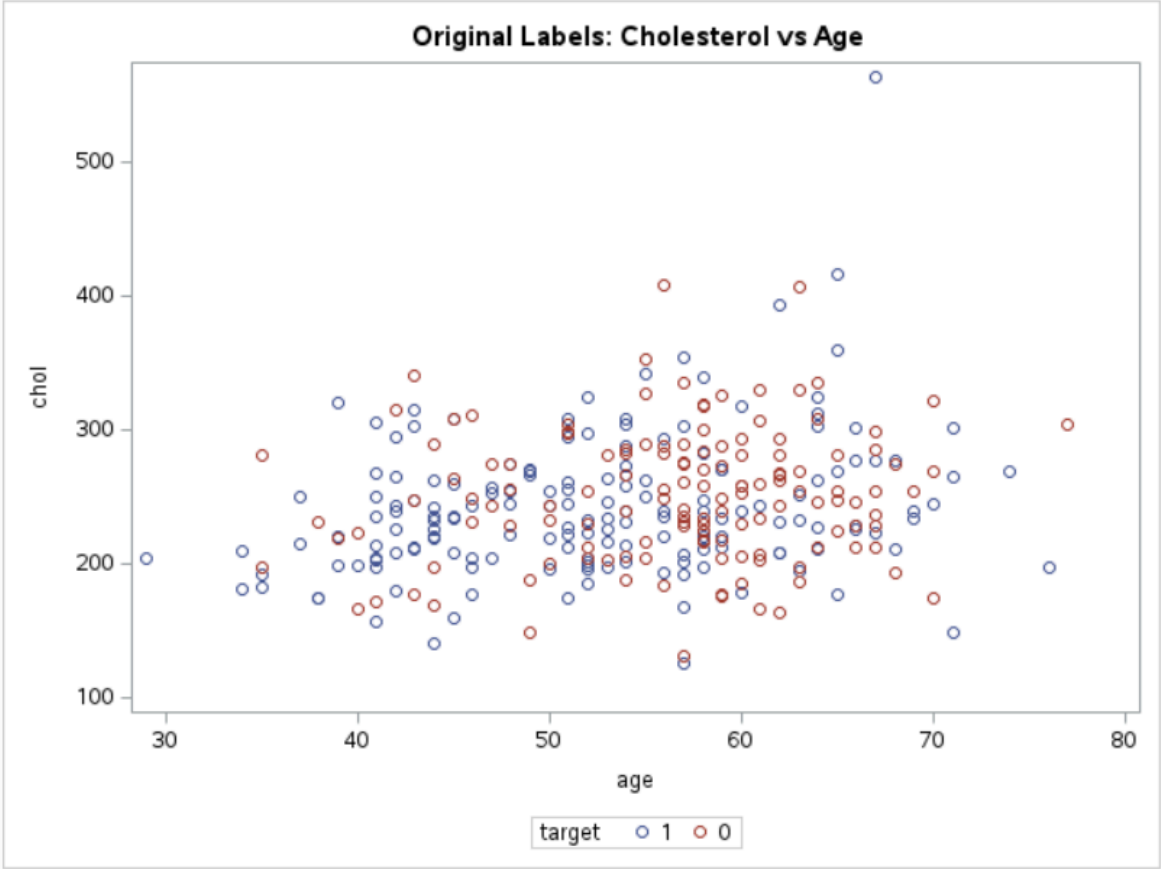
CODE/RESULT:

```
/* K-Means Clustering for K=2, 3, 4, 5 */
proc fastclus data=heart_std maxclusters=2 out=clust2 summary;
   var age trestbps chol thalach oldpeak;
run;
proc fastclus data=heart_std maxclusters=3 out=clust3 summary;
   var age trestbps chol thalach oldpeak;
run;
proc fastclus data=heart_std maxclusters=4 out=clust4 summary;
   var age trestbps chol thalach oldpeak;
run;
proc fastclus data=heart_std maxclusters=5 out=clust5 summary;
   var age trestbps chol thalach oldpeak;
run;

/*Visualization of the Cluster Labels, using K=3 */
proc sgplot data=clust3;
   scatter y=chol x=age / group=cluster;
   title "K-Means Clustering: K=3 Groups";
run;
```

```
/* Visualization of the original data labels */
proc sgplot data=heart;
   scatter y=chol x=age / group=target;
   title "Original Labels: Cholesterol vs Age";
Run;
```





ANALYSIS:

The “Criterion Based on Final Seeds” is the overall Root Mean Square Standard Deviation for the clustering solution.

- K = 2 has a value of 0.9480
- K = 3 has a value of 0.8540
- K = 4 has a value of 0.8142
- K = 5 has a value of 0.8067

A lower RMS value means the points in each cluster are more compact. Although K=5 has the lowest value, it has one cluster made of only a single observation, which suggests the fifth cluster is just removing an outlier rather than creating a meaningful group. Similarly, in K = 4, a cluster has only 5 variables, which is also very small compared to the sample size. For that reason I have decided to go with K=3 to have the lowest RMS while still having meaningful groupings.

In the graph with the original labels we can see a lot of overlapping between patients with and without heart disease.There does not appear to be a clear separation based solely on age and cholesterol.

In theK=3 clustering graph, three groups are formed. One cluster (green) captures individuals with higher standardized cholesterol and age values while the other two

clusters represent patients in the lower (red) and moderate (blue) ranges. The clusters do not perfectly match the original heart disease labels, which suggests that age and cholesterol alone are not sufficient to fully separate patients with and without heart disease.